

ROBUSTNESS UNDER DEGRADATION

How Image Distortions Break Vision Algorithms — and What Brings Them Back



Image Processing & Computer Vision — Final Project

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Dataset: COCO128 | github.com/ronivolshtein/Image-Processing-Final-Project

THE PROBLEM

Vision models are tested on clean images. Reality isn't clean.

- Cameras produce noise, darkness and motion blur — how much accuracy do vision algorithms actually lose, and how fast?
- We stress-test 4 vision tasks under 4 classical distortions, each at 4 severity levels quantified by SNR (dB).
- Then we compare two recovery strategies: classical image-processing enhancement (no training) vs. fine-tuning the object detector on distorted data.

1

How fast does each task break as SNR drops?

2

Can a matched classical filter bring accuracy back?

3

Does fine-tuning beat classical enhancement?

Deliverable: the GitHub repo — code, data, figures, and the README as the full report.

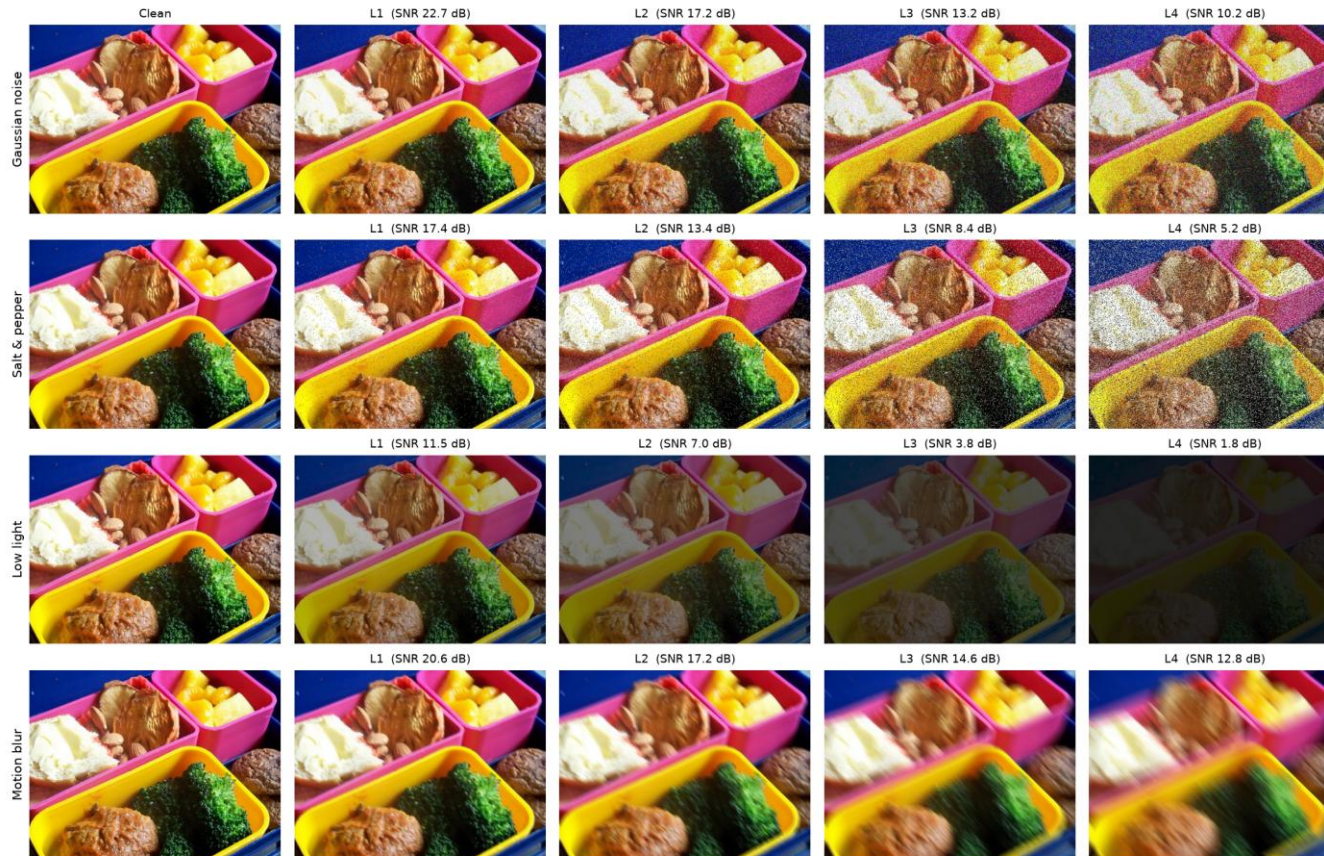
4 tasks × 4 distortions, each distortion with a matched classical fix

Component	Choice	Why
Dataset	COCO128 + COCO128-Seg	Small, public, has Ground Truth; runs on CPU
Tasks (4)	Object detection (DL) · Instance segmentation (DL) · Template matching · Sparse optical flow	High-level + low-level; classical + deep, as required
Distortions (4)	Gaussian noise · Salt & pepper · Low light · Motion blur	Geometry-preserving → original GT labels stay valid
Matched enhancement	Gaussian filter · Median filter · CLAHE · Sharpening	Each classical tool targets its distortion in theory
Recovery strategies (2)	Enhancement preprocessing · YOLO detection fine-tuning on distorted data	The two improvement approaches in the project spec

Team of 3 → per course rules: 4 tasks and 4 distortions instead of 3 and 3.

One image, four distortions, four severities — quantified as SNR

Distortion severity overview — sample 000000000009.jpg

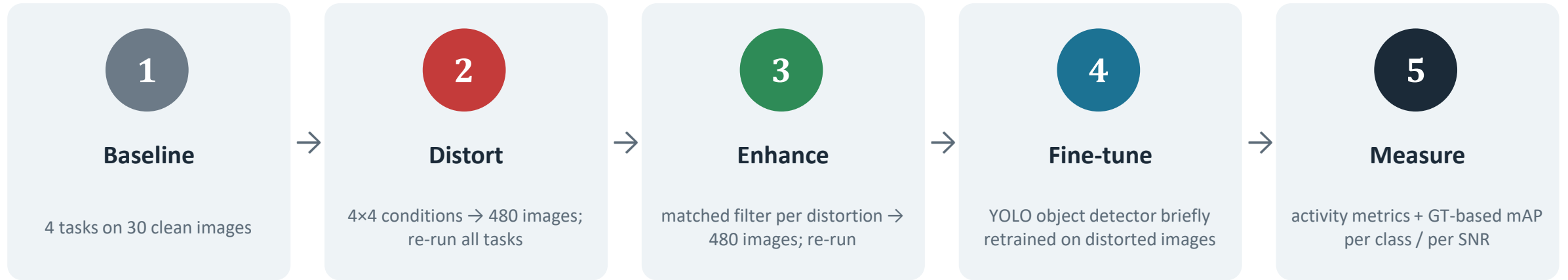


- Severity is measured, not named (averaged over the 30-image sample):

$$SNR = 10 \cdot \log_{10} \left(\frac{\text{signal power}}{\text{noise power}} \right)$$

- Gaussian σ 15→75 (22→10 dB) · Salt & pepper 2→30% (17→5 dB) · Low light (12→2 dB) · Motion blur kernel 5→35 px (19→13 dB).
- Key property: none of these move objects — so the original COCO annotations remain a valid answer key for every distorted image.

One pipeline, one central results table

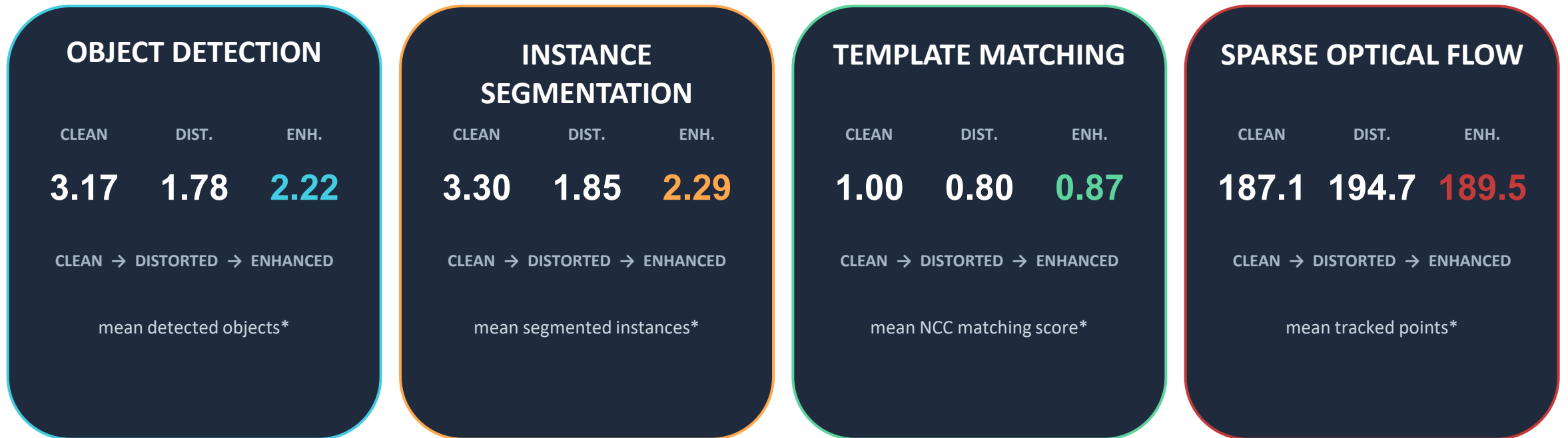


Every score is one row in one CSV. metadata_summary_base.csv: 7,950 rows — 3,570 Baseline · 3,360 Enhanced · 1,020 Fine-Tuned. GT results in map_summary.csv (1,419 rows, 33 conditions).

Fine-tuning was applied to object detection only; the segmentation model remained pretrained and was evaluated before and after enhancement.

All Four Tasks — One Summary

All tasks were re-run on clean, distorted and enhanced images. Detailed GT analysis focuses on object detection.



* Activity metrics show task behavior. Object detection is the selected GT-evaluated task: box mAP per class and per SNR.

Optical flow tracking more points under distortion reflects noise-created corners rather than improved accuracy.

Activity metrics tell you the model did something. GT tells you it was right.

Activity metrics

- Detections count, avg confidence, matching score, tracked points
- Cheap, cover all 4 tasks, show the degradation trend
- Never open the answer key — can't tell recovery from hallucination

GT-based object-detection mAP

- Every prediction checked against human-labeled Ground Truth (per class, per SNR — the course requirement)
- Possible because distortions preserve geometry: original labels stay valid
- Done for one task by design (object detection); 33 conditions × per-class

Both layers appear in the results — and they disagree in a way that became a finding.

Not all object classes start equal

0.376

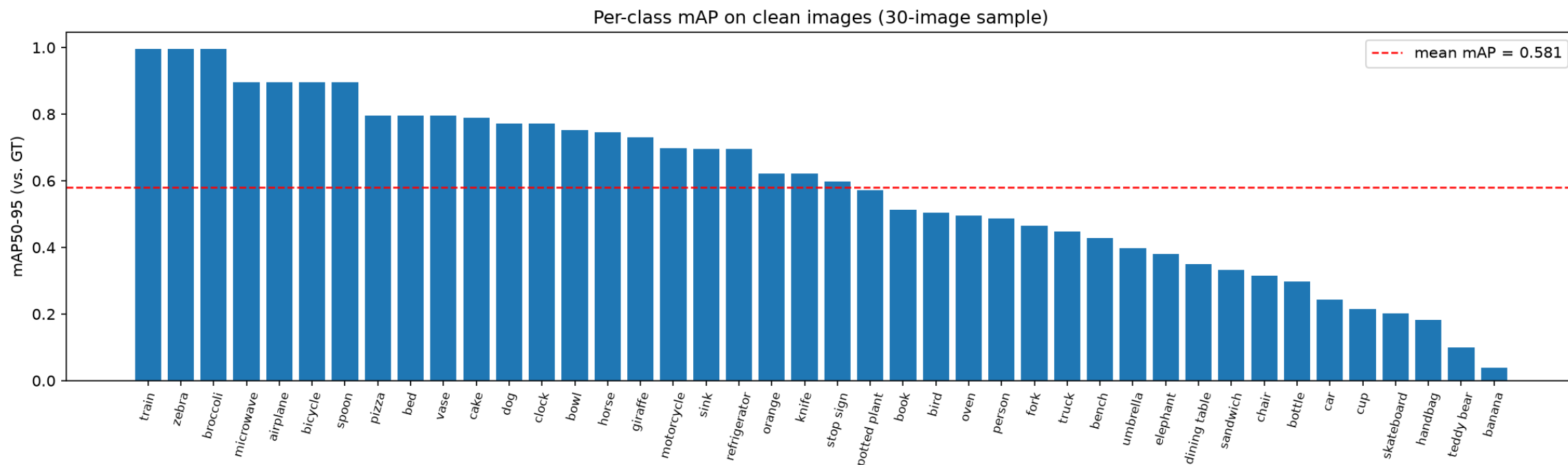
Detection mAP50-95, 128 clean images

0.284

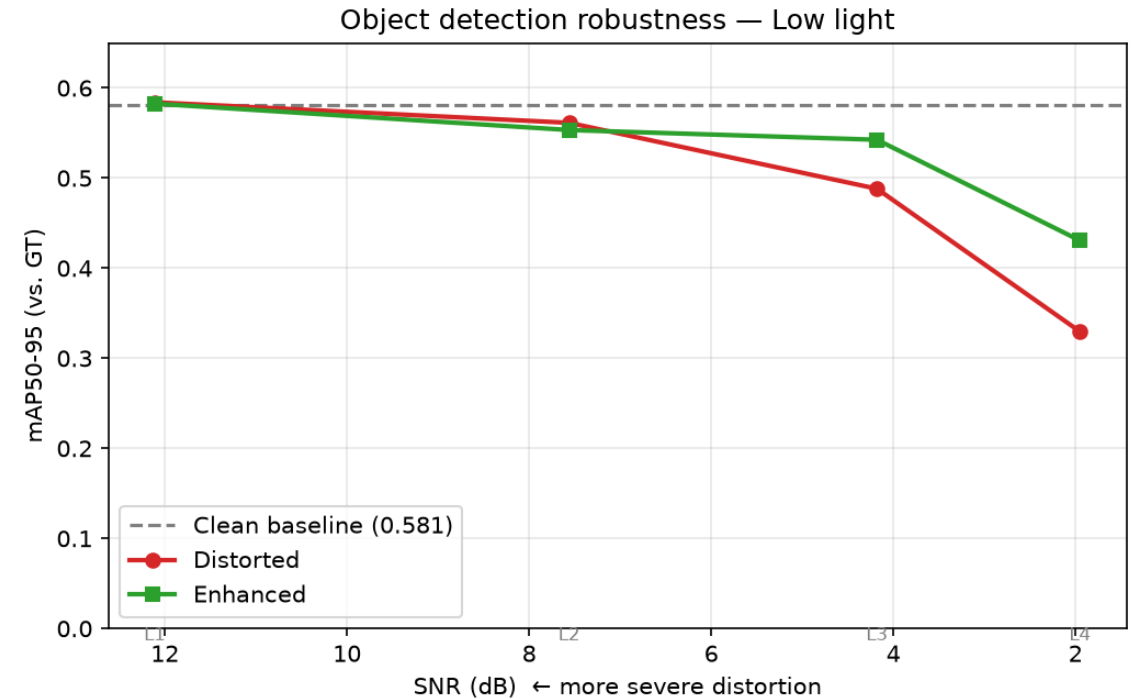
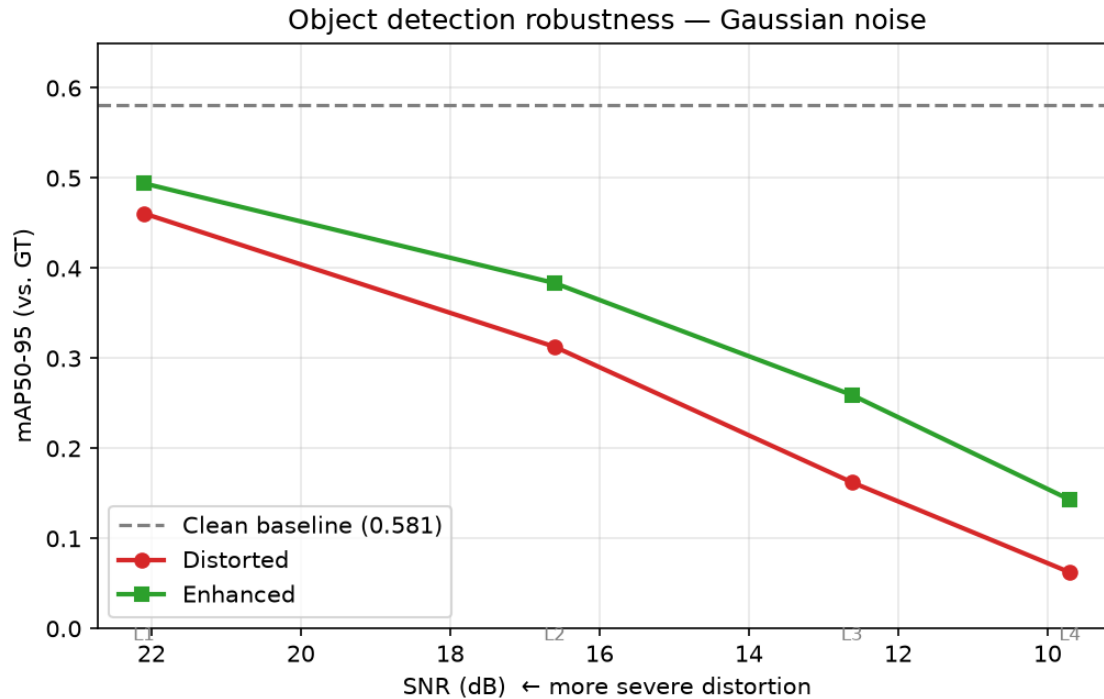
Segmentation mask mAP50-95

0.581

Detection mAP on the 30-image pipeline sample
(easier draw — all comparisons use these same 30)

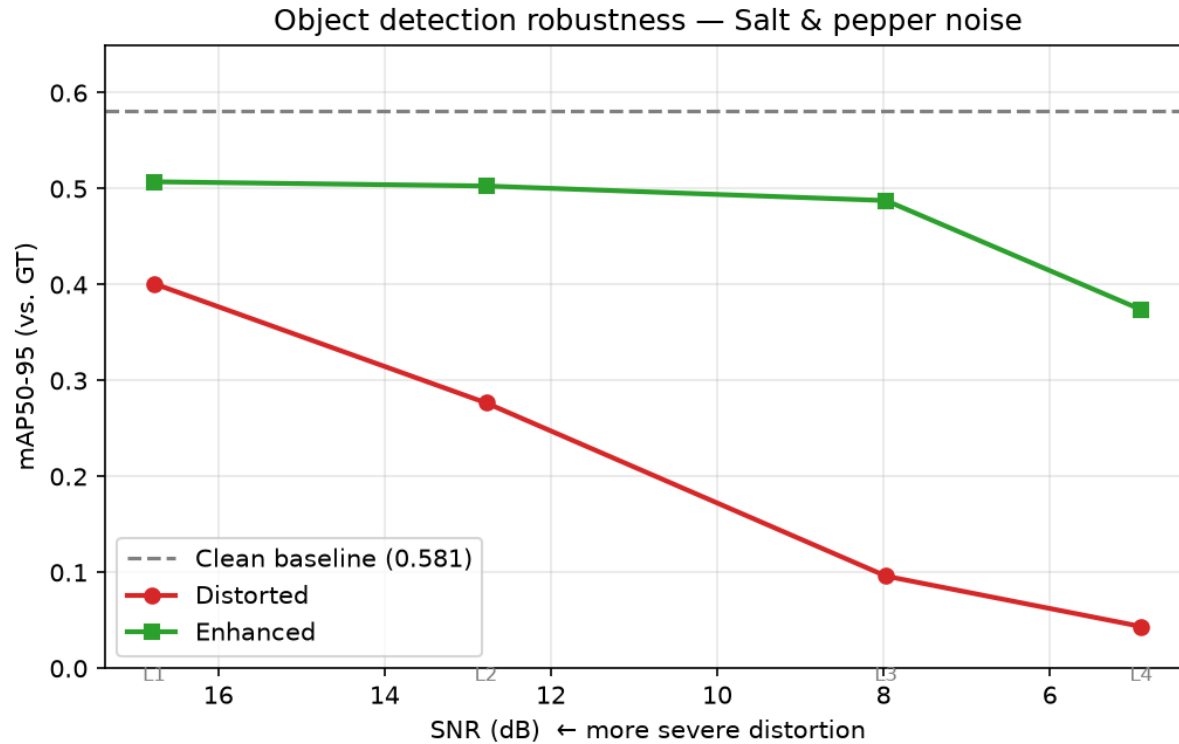


Accuracy vs. SNR: every distortion has a breaking point



- Gaussian noise: steady collapse — mAP 0.46 → 0.06 as SNR falls 22 → 10 dB.
- Low light degrades gradually at first, with the largest collapse at the most severe level.
- SNR measures pixel damage, not semantic damage: motion blur costs few dB yet destroys the edges detectors rely on.

When the tool matches the damage, dramatically yes



● **Salt & pepper + median** — near-full recovery: 0.10 → 0.49 at level 3

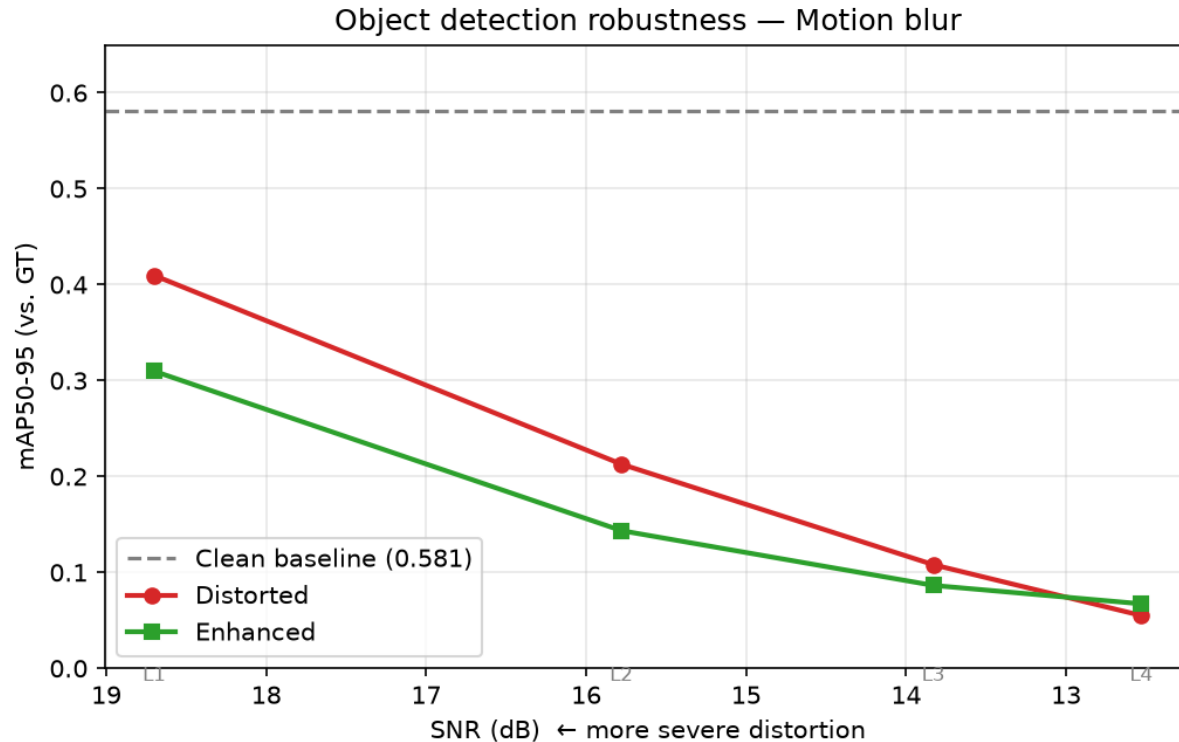
● **Low light + CLAHE** — helps exactly where damage is severe

● **Gaussian + smoothing** — moderate, consistent recovery

● **Motion blur + sharpening** — makes it WORSE — next slide

Zero training, milliseconds per image — the cheapest accuracy in the project.

Sharpening doesn't fix motion blur - it makes detection worse



0.41 → 0.31

mAP at mild blur: doing nothing vs. “helping” with unsharp masking

- Motion blur smears information along a direction; sharpening cannot un-smear it.
- It amplifies the smeared edges instead — feeding the detector confident wrong gradients.
- Real recovery needs deconvolution-style deblurring (the blur kernel is even known here).
- A negative result we retained and explained.

YOLO detections: clean → distorted → enhanced (level 3)

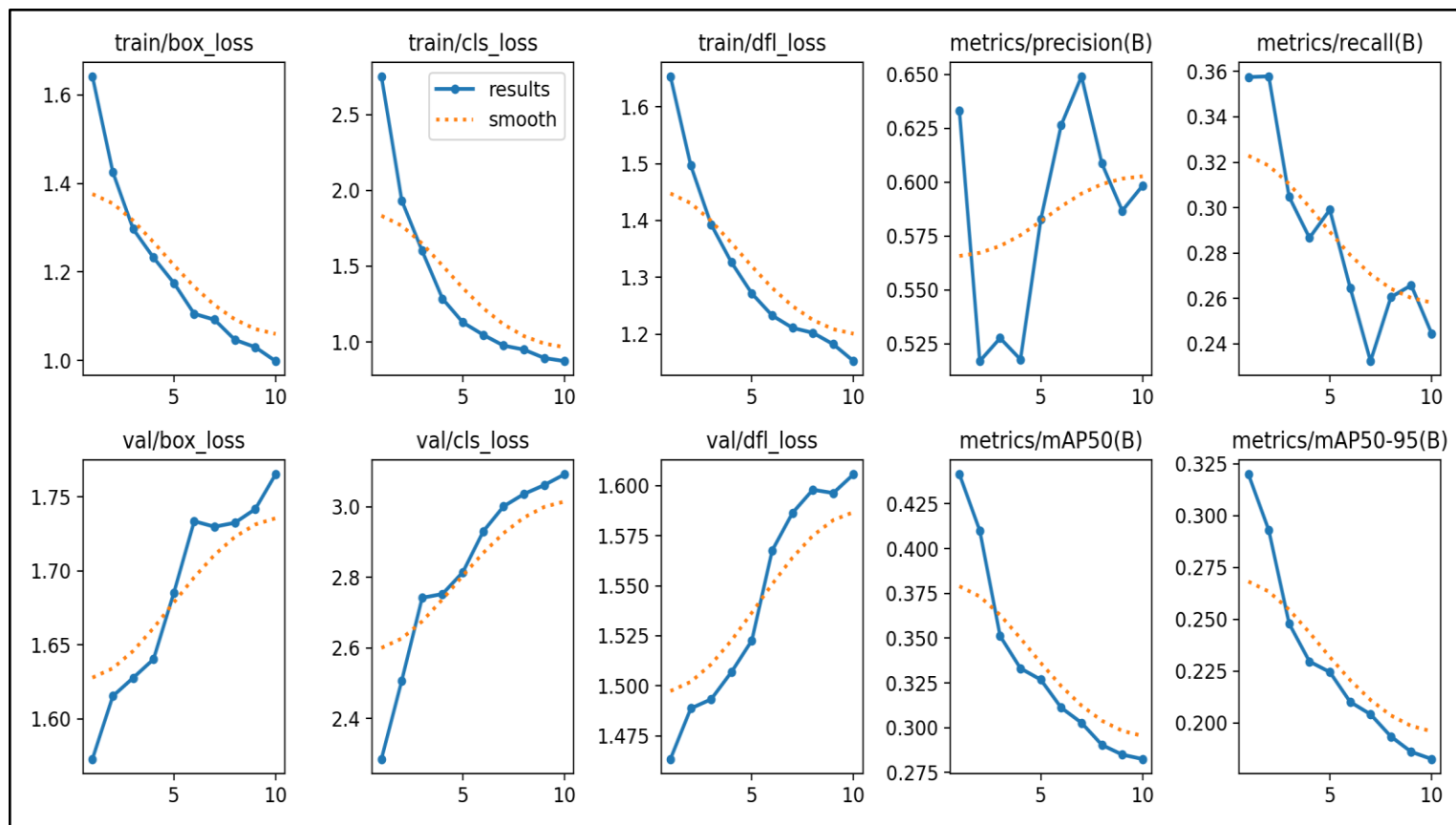
YOLO detections: clean vs. distorted vs. enhanced (level 3) — sample 000000000009.jpg



- Salt & pepper row: boxes vanish under noise — and return in full after the median filter.
- Motion blur row: after “enhancement”, still almost nothing. The curve and the picture agree.
- Gaussian row: enhancement brings boxes back, but some are low-confidence wrong classes (“dining table 0.44”) — activity without correctness, previewing the next slide.

Object-detection fine-tuning — what training showed

YOLOv8n detection · pretrained initialization · 10 epochs · 640 px · batch 8



THE GENERALIZATION GAP

Training losses decrease steadily

Validation losses rise; recall falls

mAP50-95 peaks at epoch 1: 0.320

best.pt preserves the strongest validation checkpoint

Conclusion: rapid overfitting explains why fine-tuning is treated as a small-scale extension, not the main recovery result.

Counting metrics said hero. Ground Truth said: modest.

The counting-metrics story

“+99% recovery on motion blur”

- Fine-tuned model detects 5.57 objects on clean images — the original detects 3.17 (+75% boxes everywhere)
- More detections ≠ more correct detections

The GT verdict (10 held-out images)

- Noise distortions: real but modest gains (+0.03–0.06 mAP)
- Low light & motion blur: no measurable gain
- Clean images: slightly WORSE (0.396 → 0.378) — mild catastrophic forgetting
- Evaluated on images outside the main 30-image sample

Conclusion: matched classical enhancement showed the strongest recovery in the main experiment; small-scale detection fine-tuning added modest noise robustness.

Three cases from our own data

1

Optical flow “improves” under noise 194.7 tracked points on distorted images vs. 187.1 on clean — noise manufactures fake corners for Shi-Tomasi.

2

Enhanced images “beat” clean ones Enhanced salt & pepper produced more detections than clean images; GT mAP shows recovery is real but stays below baseline.

3

The fine-tuned “super-detector” 4.58 objects detected on dark images vs. 3.17 on clean — over-detection, not skill. GT confirmed: no real low-light gain.

A robustness study needs correctness, not activity — which is why the GT layer exists.

KEY FINDINGS

1

Recovery is distortion-dependent — and matches theory

Median vs. impulse noise is the textbook pairing and nearly restores baseline; CLAHE helps at severe darkness; smoothing trades noise for detail.

2

Sharpening under motion blur is a negative result

Worse than no processing at mild–moderate blur; genuine deblurring (deconvolution) is the right tool.

3

Metrics that ignore GT overestimate — or invert — reality

Three documented cases in our own data; correctness requires the answer key.

4

Classical enhancement was the stronger result in this study

Matched classical enhancement gives the strongest recovery in the main study; small-scale detection fine-tuning adds modest noise gains at a slight cost on clean images.

Limitations: 30-image sample → per-class statistics noisy for rare classes · fine-tuning is a short, small-scale adaptation · GT evaluation covers object detection by design (one task per the project definition).

Project artifacts are in the repo.

Code · central results CSVs · all figures · README = the full report · this presentation
github.com/ronivolshtein/Image-Processing-Final-Project



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Thank you